

Enhanced detection of fluorescence fluctuations for high-throughput super-resolution imaging

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Weisong Zhao^{1,2,11}, Shiqun Zhao^{3,11}, Zhenqian Han^{1,11}, Xiangyan Ding^{1,11}, Guangwei Hu⁴, Liying Qu¹, Yuanyuan Huang¹, Xinwei Wang¹, Heng Mao⁵, Yaming Jiu⁶, Ying Hu⁷, Jiubin Tan², Xumin Ding¹✉, Liangyi Chen^{3,8}✉, Changliang Guo^{3,9,10}✉ & Haoyu Li^{1,2}✉

High-throughput super-resolution (SR) imaging is attractive for rapid and high-precision profiling in a wide range of biomedical applications. However, current SR methods require sophisticated acquisition optics and long integration times to acquire a single field of view. By exploiting the natural photophysics of fluorescence, fluctuation-based microscopy techniques can routinely break the diffraction limit without requiring additional optical components. However, their long acquisition time still poses a challenge for high-throughput imaging and the visualization of transient cellular dynamics. Here we propose super-resolution imaging based on autocorrelation with two-step deconvolution (SACD). Our method notably reduces the number of frames required by maximizing the detectable fluorescence fluctuation behaviour in each measurement. SACD requires only 20 frames to achieve a twofold improvement in lateral and axial resolution, whereas current SR optical fluctuation imaging (SOFI) needs more than 1,000 frames. With an acquisition time of ~10 min, we record SR images with 128-nm resolution over a field of view of 2.0 mm × 1.4 mm, which includes more than 2,000 cells. By applying the continuity and sparsity joint constraint, sparse deconvolution-assisted SACD enables four-dimensional imaging of live cells and events such as mitochondrial fission and fusion. Overall, as an open-sourced module, we anticipate that SACD will improve accessibility to SR imaging, thus facilitating biological studies of cells and organisms with high throughput and low cost.

In quantitative biology, microscopy-based high-throughput screening is essential to monitor the variability of biological systems and probe their heterogeneity¹. Recent advances in three-dimensional (3D) imaging resolution can contribute to minimizing uncertainties in the analysis of biological processes characterized by noise and heterogeneity.

In parallel, super-resolution (SR) microscopy techniques that rely on the blinking of single fluorophores have been developed to break the diffraction limit; these include photoactivated localization microscopy (PALM)² and stochastic optical reconstruction microscopy (STORM)³. However, the precise localization of individual fluorophores typically

A full list of affiliations appears at the end of the paper. ✉e-mail: xuminding@hit.edu.cn; lychen@pku.edu.cn; changelguo@gmail.com; lihaoyu@hit.edu.cn

requires tens of thousands of frames to generate one single SR image, which inherently limits the throughput⁴. Therefore, although PALM/STORM has been reported with live cells^{5–7}, the high illumination power ($\sim 10 \text{ kW cm}^{-2}$)⁸, long exposure time ($> 2 \text{ s}$)⁹ and the need for photochemical environments that promote long dark states to reduce photobleaching⁹ prevent PALM/STORM techniques from becoming a customary live-cell imaging method¹⁰ with high throughput.

The localization mechanism essentially restricts the maximum density of emitting fluorophores in one frame¹¹. As a solution to this, fluorescence fluctuation-based SR microscopy techniques^{12–15} are used. These rely on independent stochastic blinking/fluctuations of many emitters rather than single ones. Remarkably, without the need for sophisticated acquisition devices, SR optical fluctuation imaging (SOFI)¹² calculates the n th-order autocorrelation cumulants, yielding $\sqrt{n}$ -fold improvements in resolution in all three dimensions. Thanks to the relaxation on using the blinking of single molecules and on the requirement for stringent photochemical environments^{16,17}, SOFI is faster than PALM/STORM and is suitable for a range of live-cell SR imaging applications, such as observing the dynamics of focal adhesion points¹⁸, clathrin-coated pits (CCPs)¹⁹ and actin²⁰. However, the temporal resolution and throughput of SOFI remains limited, because hundreds to thousands of frames are typically needed to reconstruct one SR image¹⁸.

In this work, we develop ‘super-resolution imaging based on autocorrelation with two-step deconvolution’ (SACD), which minimizes the number of frames required for fluctuation-based imaging modalities. By maximizing the detectable fluorescence fluctuation behaviour in each measurement, SACD only requires 20 frames to achieve a twofold improvement in lateral and axial resolutions, unlike conventional SOFI, which requires $\sim 1,000$ frames to attain the same improvement. To achieve fully automatic calculations, we also use Fourier ring correlation (FRC)^{21,22} to non-biasedly determine the iteration times of deconvolution. We demonstrate high-throughput SR imaging with a spatial resolution of $\sim 128 \text{ nm}$ over an $\sim 2 \text{ mm} \times 1.4 \text{ mm}$ area within $\sim 10 \text{ min}$, a throughput that has been hard to achieve previously²³. In addition, we adapt SACD to reconstruct images under low signal-to-noise-ratio (SNR) conditions by constraining the sparsity and continuity of prior knowledge²⁴ in the post-deconvolution step. This solution addresses the problem of deteriorating SNR and cell viability during long-term live-cell SR imaging due to photobleaching and phototoxicity²⁴. We capture rapid dynamics of mitochondrial membrane structures, such as fission and fusion. Overall, we anticipate that SACD will extend the capabilities of 4D live-cell studies, allowing SR imaging of processes with high temporal resolution and long recording times. To facilitate its dissemination and future applications, we provide an easy-to-use library and Fiji/ImageJ plugin, offering an add-on SR feature for any off-the-shelf commercialized microscopes or in-house custom-built systems.

Results

SACD model development

Despite the fact that SOFI theoretically has unlimited spatial resolution, three fundamental issues limit its high-order ($n > 2$) achievements in practice: (1) its nonlinear response to brightness and blinking heterogeneities²⁰, (2) potential reconstruction artefacts²⁵ and (3) the requirement for long sequences¹⁸. Reconstruction from extended image sequences often leads to motion artefacts blurred by highly mobile subcellular structures in live cells. Here we focus on second-order ($n = 2$) SOFI over the higher-order one.

In principle, the autocorrelation cumulant can eliminate any noise uncorrelated with time¹². However, the statistical uncertainty arising from limited datasets may dramatically affect continuity and homogeneity²⁶. Hence, SOFI generally requires at least hundreds of raw images to preserve structural integrity. Predictably, the magnitude of the statistical uncertainty is highly related to the quantity and accuracy of the

raw data. When considering the accuracy only, two properties of each measurement influence the magnitude of the statistical error: (1) the captured effective on/off contrast ratio and (2) the SNR condition^{17,19}. In SOFI, the sample is seen as being composed of N independently fluctuating emitters (Extended Data Fig. 1a and Methods)¹². Besides in-focus fluorescent molecules, the image is corrupted by out-of-focus and cytosol fluorescence signals that may not remain in stationary equilibrium (Extended Data Fig. 1a and Methods). These aberrant fluorescence signals compromise the effective on/off contrast ratio, while the sensor's readout noise, dark current and shot noise reduce the image SNR. Accordingly, more raw frames are required to increase the accuracy and quality of the SR images^{12,18}, which leads to an ultimate temporal limit to fluctuation-based SR microscopy.

In this work, we apply Richardson–Lucy (RL) deconvolution^{27,28} on the raw image stack as a preprocessing step (Fig. 1a), aiming to filter the out-of-focus and cytosol background. Such a procedure improves the effective on/off contrast ratio and reduces the random pixel-level fluctuations induced by noise, while maintaining linear responses across frames (Extended Data Fig. 2; details are provided in Supplementary Note 1). To avoid destroying the independence of fluorescence fluctuation due to excessive RL deconvolution, we cautiously stopped the iteration at an early stage (Methods and Extended Data Fig. 1c). This preprocessing step highlights the structures in raw images, which effectively increases the accuracy of each measurement. Based on this preprocessing, 20 frames are workable for the next autocorrelation cumulant calculation with a Fourier interpolation²⁹ (Fig. 1a and Methods). We then added a postprocessing step once again to deconvolve the second-order SOFI images (Fig. 1a), to further extend the resolutions. Notably, this deconvolution step also provides the basis for correcting the squared brightness caused by the second-order cumulant, without cancelling the improvement in resolution³⁰. Accordingly, we applied a square root on the reconstruction as a last step to directly linearize the brightness (Fig. 1a; details are provided in the Methods). Furthermore, to automatically adjust the tunable parameters, we introduced FRC^{21,22} as a quantitative resolution metric to assist in the process (Extended Data Fig. 1b and Methods). Specifically, the FRC was used to determine the optimal iteration times and estimate the effective point-spread function (PSF) after the autocorrelation cumulant calculation²² (Extended Data Fig. 1b).

We systematically compared the performances of SACD against other state-of-the-art fluctuation-based SR methods^{31–33} in simulations (Methods, Extended Data Figs. 3 and 4 and Supplementary Note 2). Both visual and quantitative examinations showed that SACD can decode SR details from low-contrast fluctuations with minimum input. With these tests, we found that the measured accuracy in each frame is the essential factor for fluctuation-based SR microscopy, rather than the quantity (1000 frames) considered previously. We also replaced the pre-RL deconvolution with different preprocessing operations, and they were not comparable to our current solution (Supplementary Note 3).

Cross-validation with other SR techniques

We first tested SOFI and SACD on a spinning-disk confocal (SD-confocal) microscope (Methods). SACD consistently achieved superior image quality and resolution enhancement in comparison to the 1,000-frame SOFI (Extended Data Fig. 5; for details see Supplementary Note 5). We then examined the reliability of SACD in resolving fine structure details and its broad applicability to different imaging modalities. A custom-built structured illumination microscope (SIM; Methods)³⁴ was used to acquire 2D-SIM images of microtubules labelled with quantum dots 525 (QD₅₂₅), which can provide in situ reference images for SACD (Fig. 1b). As shown in Fig. 1c, we found that the 20-frame SACD successfully resolved the intertwining filament structures, identically to the corresponding 2D-SIM image; the 1,000-frame SACD gave an almost identical result (Extended Data Fig. 6e), indicating that

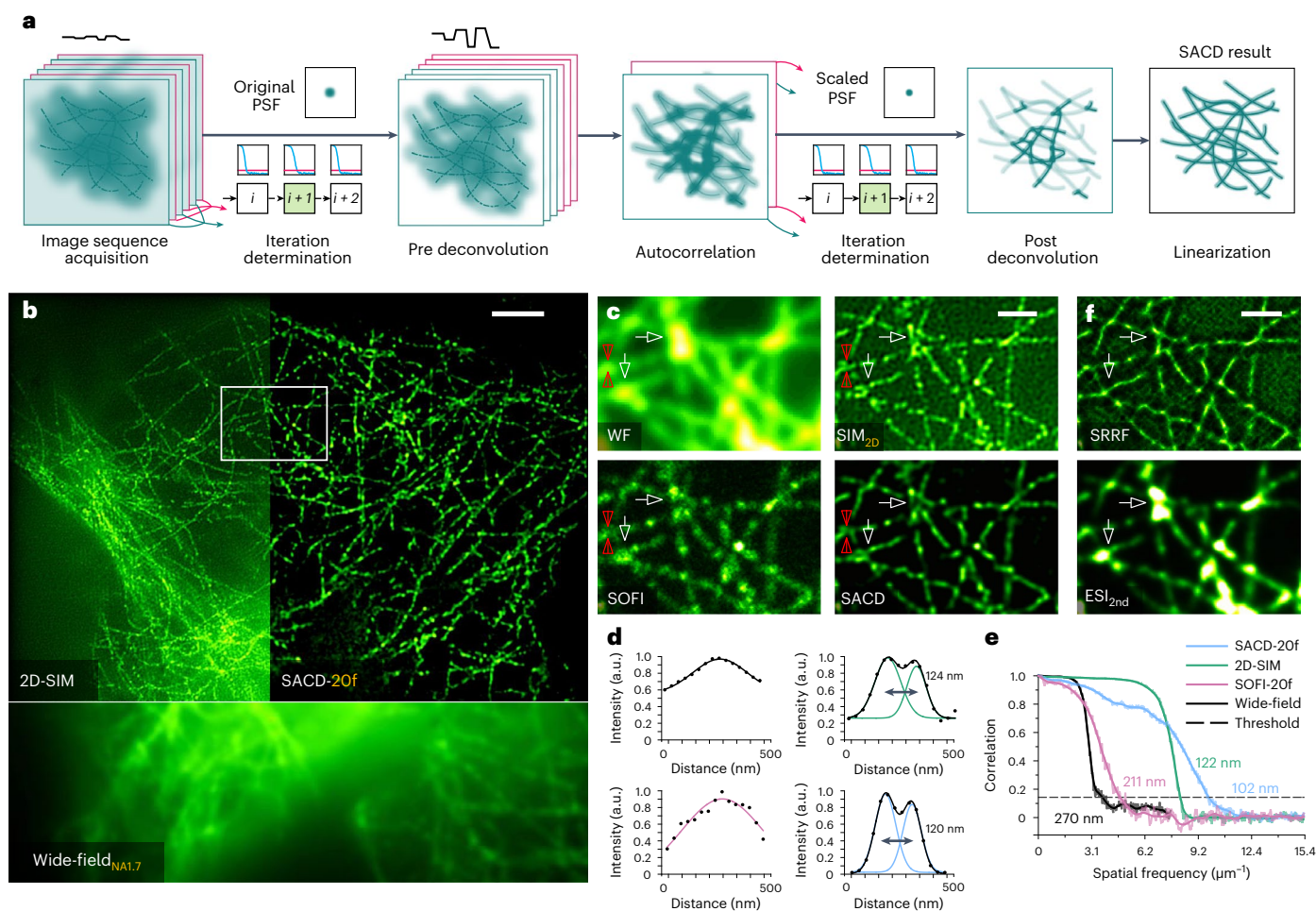


Fig. 1 | Workflow and cross-validation of SACC. **a**, Reconstruction of the workflow of SACC. **b**, Microtubule filaments in a COS-7 cell labelled with QD₅₂₅ imaged by wide-field microscopy (bottom), 2D-SIM (left) and 20-frame SACC (right). Scale bar, 3 μm . **c**, Zoomed views from the white box in **b**. Top left: wide-field image (averaged by 20 frames). Top right: 2D-SIM image. Bottom left: 20-frame SOFI image. Bottom right: 20-frame SACC image. Scale bar, 1 μm . **d**, Intensity profiles and multiple Gaussian fitting of raw wide-field (top left),

2D-SIM (top right), 20-frame SOFI (bottom left) and 20-frame SACC (bottom right) for the structures of microtubule filaments indicated by the red arrows in **c**. The numbers indicate the distance between peaks. **e**, FRC analysis of the reconstructed images. **f**, Zoomed views from the white box in **b** of 20-frame SRRF with temporal radially average (top) and 20-frame second-order ESI (bottom). Scale bar, 1 μm . Experiments were repeated ten times independently, with similar results.

the 20-frame input is sufficient. Benefiting from the improved axial resolution, SACC eliminated the out-of-focus light, thus offering a higher contrast than 2D-SIM (Fig. 1b). Analysis of the peak-to-peak separations of adjacent microtubule filaments indicated an achievable resolution of ~120 nm for SACC and ~124 nm for 2D-SIM (Fig. 1d). The resolutions in the wide-field (270 nm), SOFI (211 nm), SIM (122 nm) and SACC (102 nm) images were further confirmed by FRC analysis (Fig. 1e). Note that, because of the influence of nonlinearity on the resolution estimation, linearization is always used before FRC calculations (Extended Data Fig. 6b).

Furthermore, by taking advantage of the SIM image as a dependable reference, we benchmarked the superior performances of SACC with three other fluctuation-based methods (Fig. 1f): SOFI, SR radial fluctuations (SRRF)¹³ and entropy-based SR imaging (ESI)¹⁵. As indicated by the white arrows in Fig. 1, SOFI, SRRF and ESI failed to resolve fine structures due to the lack of raw images. In the SACC and SIM images, the intertwining filaments could be determined, and SACC also demonstrated its superior capability for achieving background-free and contrast-enhanced results. We also compared the modified SOFI methods using the same dataset, and they achieved acceptable reconstruction only with the 1,000-frame configuration (Extended Data Fig. 6a–e; details are provided in Supplementary Note 9).

3D-resolution doubling by SACC

To evaluate the resolution enhancement of SACC in all three dimensions, we extended the experiments to volumetric imaging. In a preliminary investigation, we measured the 3D PSF for a sample of QD₅₂₅ randomly dispersed on a substrate, captured by a standard SD-confocal microscope (Fig. 2a and Methods). We extracted the lateral/axial full-width at half-maxima (FWHMs) of this point source as 325/655 nm and 296/510 nm in SOFI and 135/334 nm in SACC (all with 20 frames), averaged for SD-confocal and reconstructed for SOFI and SACC. The resulting FWHMs indicate that SACC effectively achieved a twofold resolution enhancement in all three dimensions (Supplementary Fig. 1). Compared to SOFI, SACC improved the homogeneity of the PSF, even when using a limited dataset (20 frames). Next, we demonstrated the 3D SR ability of SACC by imaging the microtubule cytoskeleton (Extended Data Fig. 7a). As expected with a SD-confocal microscope, we found that the fine features are blurred in all directions, especially along the axial direction (Extended Data Fig. 7b,c). In contrast, SACC clearly showed the microtubule network in all three dimensions (Extended Data Fig. 7b,c).

Because SACC requires no specific optical configuration, we can directly use the full sCMOS (scientific Complementary Metal–Oxide–Semiconductor) camera field of view (FOV, ~130 × 130 μm^2 , Fig. 2b,

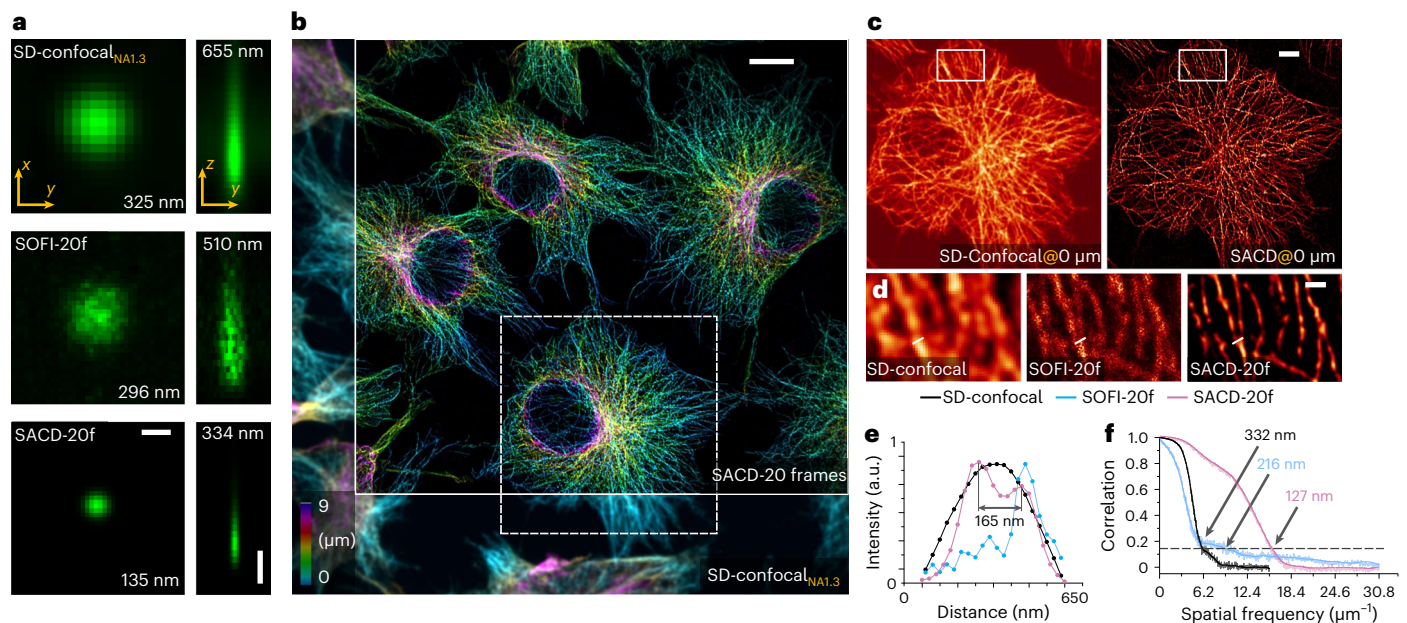


Fig. 2 | Validation for the 3D-resolution doubling of SACD and large FOV volumetric imaging. **a**, From top to bottom: maximum intensity projections (left) and vertical sections (right) of a single QD₅₂₅ imaged by SD-confocal (top), SOFI with 20 frames (middle) and SACD with 20 frames. The numbers indicate the lateral (left) or axial (right) resolutions. Scale bars, 500 nm. **b**, Colour-coded, 3D distributions of microtubule filaments in COS-7 cells labelled with QD₅₂₅ obtained with SD-confocal (bottom) and SACD (top) (Supplementary Video 1).

Scale bar, 10 μm . **c**, Zoomed horizontal section views from the white dashed box in **b**. Scale bar, 5 μm . **d**, Enlarged regions from the white boxes in **c**. Scale bar, 1 μm . **e**, Intensity profiles of raw SD-confocal (green), 20-frame SOFI (blue) and 20-frame SACD (red) for the structures of microtubule filaments indicated by the white lines in **d**. The numbers indicate the distance between peaks. **f**, FRC analysis of the reconstructed images. The dashed line represents the FRC threshold. Experiments were repeated five times independently, with similar results.

Supplementary Fig. 2 and Supplementary Video 1) to implement 3D SR imaging. Existing commercial SIM set-ups are mostly limited to a FOV equivalent to a third to a quarter of the camera ($30 \times 30 \mu\text{m}^2$ to $60 \times 60 \mu\text{m}^2$, shown by a white dashed box in Fig. 2b)³⁵. Taking advantage of this, we simultaneously imaged multiple COS-7 cells at a depth of 9 μm with the doubled resolutions (Fig. 2c–f). We found that the 20-frame SOFI image suffered from limited quality, and its continuity appeared insufficient (Fig. 2d). In comparison, the 20-frame SACD (Fig. 2d) consistently performed a reliable resolution enhancement, distinguishing a 165-nm distance for the microtubule filaments (Fig. 2e). FRC analysis (Fig. 2f) also revealed that SACD (127 nm, in red) notably outperformed 20-frame SOFI (216 nm, in blue). This superiority emerged for all spatial frequencies.

Direct high-throughput SR imaging

With no need for additional optical devices or specific photoactivation control, SACD can directly enable an add-on SR feature for various fluorescence imaging systems. Beyond that, the drastic improvement in the efficiency of SACD permits SR imaging of orders-of-magnitude more cells and FOVs per unit time, immediately lending itself to high-throughput imaging. In our experiments we used an automated acquisition protocol in a commercial SD-confocal microscope to image microtubules on a millimetre-scale ($2.0 \text{ mm} \times 1.4 \text{ mm}$) area with 32×22 partly overlapping FOVs ($\sim 66.6 \mu\text{m} \times 66.6 \mu\text{m}$ each; details are provided in the Methods). The imaging time was within ~ 10 min for SACD, whereas SOFI required ~ 8 h to achieve a similar imaging performance.

In Fig. 3a and Supplementary Video 2 we show that we identified 704 sub-FOVs in total, and each sub-FOV was captured with 20 frames. We then stitched all the resulting super-resolved sub-FOVs into one single, large FOV containing more than 2,000 cells. Stitching the reconstructed images yielded a single SR image containing ~ 2.4 gigapixels (32.5-nm pixel size), spanning almost five orders of magnitude in spatial scale. Zooming into the randomly chosen sub-FOVs, the SACD images

show the microtubule filament structures clearly, whereas neither the SD-confocal images nor 20-frame SOFI provided such small-scale information (Fig. 3d and Supplementary Fig. 3a–c).

To quantify the resolution improvements over the entire area, we measured the FRC resolutions of SD-confocal and SACD for each sub-FOV, and assembled them into a resolution map (Fig. 3b). SACD consistently provided improved effective resolution compared with SD-confocal, from ~ 442 nm to ~ 128 nm (Fig. 3c and Supplementary Fig. 3d,e) over the entire large FOV in Fig. 3a. In addition, two-peak analysis confirmed the FRC measurements, enabling a 163-nm distance of the filaments to be determined (Fig. 3d, inset). Ultimately, all these analyses confirm the ability of SACD to facilitate effortless high-throughput SR imaging.

Our SACD concept can also be applied to other fluctuation-based techniques, offering further imaging speed and quality enhancements (Extended Data Fig. 8 and Supplementary Video 3; details are provided in Supplementary Note 6). We also imaged two nucleolar proteins in live cells (Extended Data Fig. 9) to explore its broad applicability for different samples.

Rapid 4D live-cell SR imaging by Sparse-SACD

To maximize the utilization of accessible photon flux, we introduced our previously developed Sparse deconvolution²⁴, splitting it into a pre-background subtraction and a post-deconvolution constrained by using continuity and sparsity penalties in the $xy-t(z)$ axes (Methods, Fig. 4a and Supplementary Note 7). Throughout 20 s of consecutive SR imaging, we found that the SNR in the raw data decreased rapidly (Fig. 4b,c and Supplementary Video 4). Although enhanced substantially compared with SOFI (Fig. 4c), this SNR reduction still led to a decrease in continuity and fidelity (Fig. 4c). Applying the continuity and sparsity joint constraint, the Sparse deconvolution-assisted SACD largely suppressed potential artefacts, yielding high-fidelity continuous filaments (Fig. 4c and Supplementary Fig. 4).

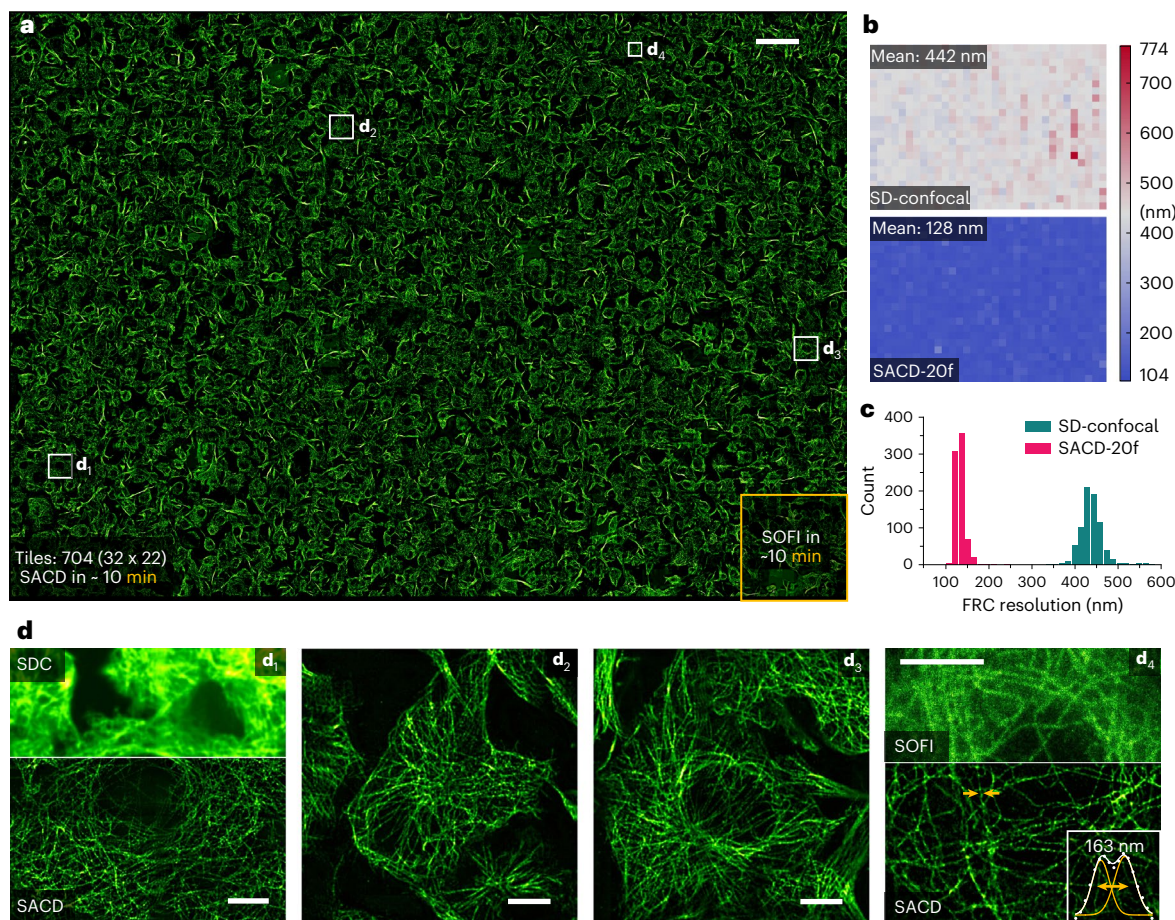


Fig. 3 | High-throughput SR imaging with SADC. **a**, Application of SADC (with 20 frames) to high-throughput SR imaging of an ~ 2.0 mm $\times$ 1.4 mm area containing more than 2,000 cells. Microtubules are identified in COS-7 cells labelled with QD₆₀₅ (Supplementary Video 2). Scale bar, 0.1 mm. **b**, FRC resolution maps over the entire FOV of the SD-confocal image (top) and SADC image (bottom). **c**, FRC resolution distributions of SD-confocal (green) and SADC

(magenta). **d**, Zoomed views of the white boxed regions in **a**. Scale bars, 10 μ m. Inset: intensity profiles and multiple Gaussian fitting of SADC for the structures of microtubule filaments indicated by yellow arrows. The number indicates the distance between peaks. SDC, SD-confocal. Experiments were repeated three times independently, with similar results.

We next applied Sparse-SADC to capture live COS-7 cells (labelled with SkyScan-S-TOM20) over 10 min with a 40-s interval (Fig. 4d, Extended Data Fig. 10 and Supplementary Video 5). The mitochondria were clearly resolved as hollow organelles, and their fine changes in morphology during fission (Fig. 4e, cyan arrows) and fusion (Fig. 4e, magenta arrows) were caught in several sequential SR recordings across minutes. Under such live-cell conditions, the previous modified SOFI methods produced severely aberrated SR images, especially at the end imaging stage (Extended Data Fig. 6d,e; for details see Supplementary Note 9). Beyond that, we also applied SADC to the challenge of 4D SR imaging in whole live cells (Fig. 4f and Supplementary Video 6). Compared to the confocal system, the visibility of fine structures at depth was greatly improved, and the whole-cell mitochondrial network could be observed well over 10 min.

Discussion

In an abstract sense, individually fluctuating molecules distributed in a strong background are like ‘shining stars in the mist’. Our core intention was to dispel this ‘mist’ before the cumulant calculation, to enable high-quality SR under real physiological conditions. We thus developed an easy-to-use algorithm, SADC, for efficient high-throughput and live-cell SR imaging. By fully leveraging fluorescence fluctuation behaviour from minimal datasets, our SADC breaks the frame-number limitation of conventional SOFI, reducing the requisite frames by orders of magnitude. Utilizing our method, only 20 frames are employed instead of hundreds to thousands, offering a rapid imaging capability for gaining

insight into biological structures at the scale of ~ 100 nm. SADC was then directly applied to a high-throughput SD-confocal system, achieving a stable ~ 128 -nm resolution over the entire centimetre-scale FOV. This demonstrates that SADC may be beneficial for high-throughput screening, for example, for particle analyses³⁵ or drug treatments⁴.

According to the systematic analyses in Supplementary Note 4, the second-order SADC shows fast convergence at 20 frames, and its performance is insensitive to changes in fluctuation behaviour and label density. This suggests that the use of second-order SADC is feasible with a variety of different probes, such as organic dyes, fluorescent proteins or nanoparticles³⁶. In contrast, although we found that SADC can notably improve the efficiency of third-order and fourth-order cumulant calculations, the use of higher-order statistics requires more frames (Supplementary Note 4). Unlike the second-order case, the convergence of high-order SADC appears highly sensitive to fluctuation behaviour, which it inherits from SOFI. Thus, further investigations may focus on reducing the frames necessary for high-order SADC reconstruction. It is likely that this will further extend the spatial resolution and even facilitate quantitative studies, such as for the extraction of molecule density on cell membranes³⁷. When applied to a large area of flat structures, 20-frame SADC may exhibit insufficient continuity and require more frame averaging to minimize intensity fluctuations (Supplementary Note 8). To ensure image quality, Sparse deconvolution can be applied as a post-deconvolution step (Supplementary Note 8). Finally, it is also interesting to note that our SADC framework

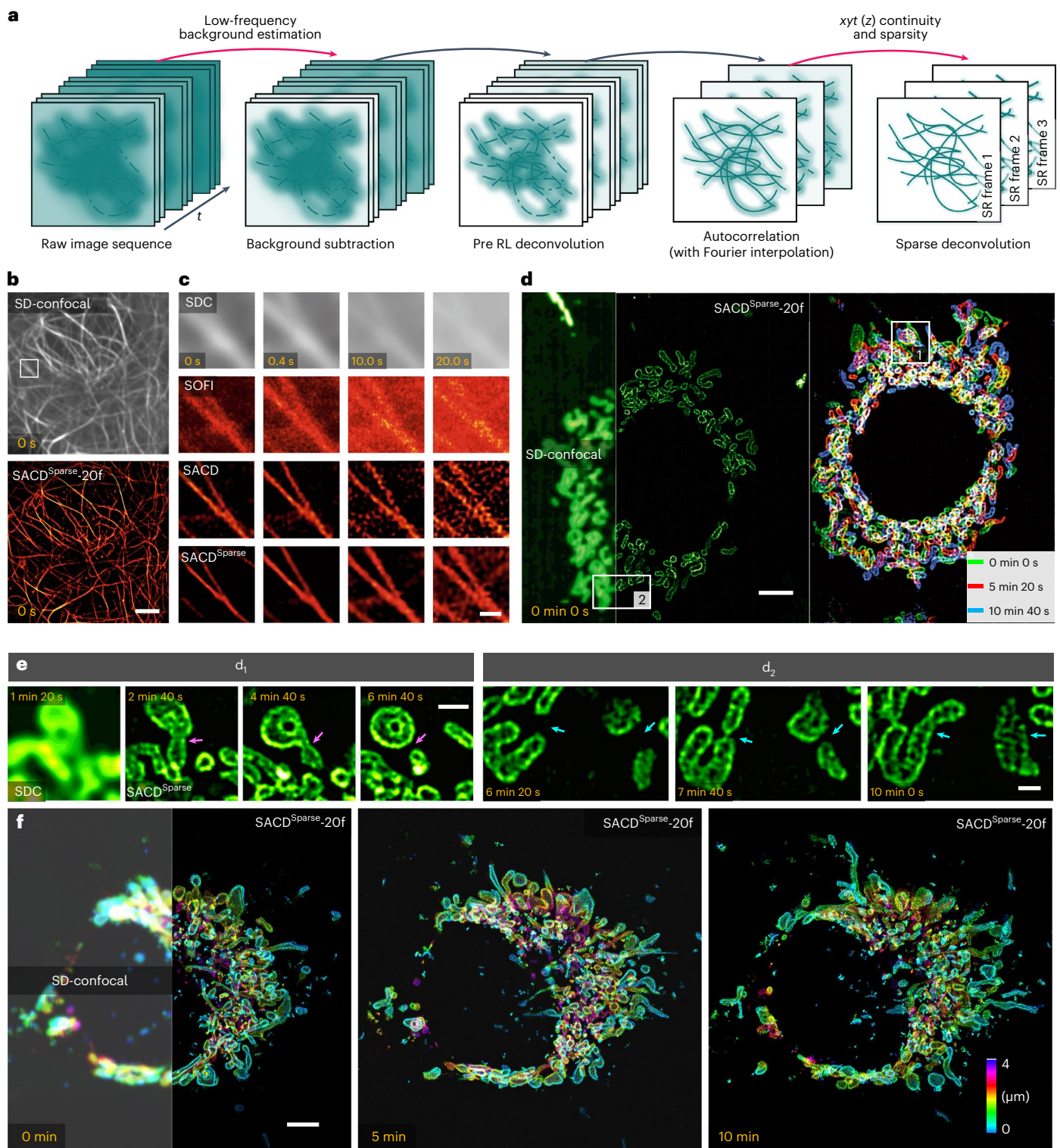


Fig. 4 | 4D live-cell SR imaging using Sparse-SACD. a, Reconstruction of the workflow of Sparse-SACD. **b**, A representative live COS-7 cell labelled with MAP4-Skylan-S imaged at 37 °C by SD-confocal (top) and Sparse-SACD ('SACD^{Sparse}') with 20 frames (bottom) (Supplementary Video 4). Scale bar, 5 μm . **c**, From top to bottom: representative montages of enlarged regions enclosed by the white box in **b** for SD-confocal (labelled SDC), SOFI, SDC and Sparse-SACD with 20 frames. Scale bar, 1 μm . **d**, From left to right: a representative live COS-7 cell labelled with Skylan-S-TOM20 imaged at 37 °C by SD-confocal (left) and Sparse-SACD (middle)

at the first time point, and the colour-coded temporal projection of Sparse-SACD (right) (Supplementary Video 5). Scale bar, 5 μm . **e**, Magnified montages of the corresponding white boxed regions in **d**. The first one is the SD-confocal result ('SDC') and the others are Sparse-SACD results. Scale bars, 1 μm . **f**, 4D imaging of a live COS-7 cell labelled with Skylan-S-TOM20 at 37 °C using SD-confocal (left) and Sparse-SACD (right) (Supplementary Video 6). Scale bar, 5 μm . Experiments were repeated five times independently, with similar results.

is feasible for analysing a wide range of densities in single-molecule localization microscopy (SMLM) data¹¹, enabling superior performance with high-density images (Supplementary Fig. 5).

We have shown that the concept of SADC can directly achieve resolution improvement on a commercial system. We anticipate that the complete SADC toolkit will become a routine tool for biologists to dissect intricate and transient dynamics, and it may facilitate microscopy-based screening with previously inaccessible high throughput, SR and low cost.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41566-023-01234-9>.

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¹Innovation Photonics and Imaging Center, School of Instrumentation Science and Engineering, Harbin Institute of Technology, Harbin, China.

²Key Laboratory of Ultra-precision Intelligent Instrumentation of Ministry of Industry and Information Technology, Harbin Institute of Technology, Harbin, China. ³State Key Laboratory of Membrane Biology, Beijing Key Laboratory of Cardiometabolic Molecular Medicine, Institute of Molecular Medicine, National Biomedical Imaging Center, School of Future Technology, Peking University, Beijing, China. ⁴School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore, Singapore. ⁵School of Mathematical Sciences, Peking University, Beijing, China. ⁶Unit of Cell Biology and Imaging Study of Pathogen Host Interaction, The Center for Microbes, Development and Health, Key Laboratory of Molecular Virology and Immunology, Institut Pasteur of Shanghai, Chinese Academy of Sciences, Shanghai, China. ⁷School of Life Science and Technology, Harbin Institute of Technology, Harbin, China. ⁸PKU-IDG/McGovern Institute for Brain Research, Beijing, China. ⁹Jingjinji National Center of Technology Innovation, Beijing, China.

¹⁰Beijing Institute of Collaborative Innovation, Beijing, China. ¹¹These authors contributed equally: Weisong Zhao, Shiqun Zhao, Zhenqian Han, Xiangyan Ding.

✉ e-mail: xuminding@hit.edu.cn; lychen@pku.edu.cn; changlguo@gmail.com; lihaoyu@hit.edu.cn

Methods

SACD reconstruction

Fluctuations-based SR imaging. For simplicity, a sample to be imaged is usually regarded as being composed of N single, independently fluctuating emitters¹², located at position r_k . These emitters are considered to have an individual time-dependent fluctuating molecular brightness, and the fluorescence signal at position r and time t can be given as

$$F(r, t) = \sum_{k=1}^N h(r - r_k) \times c_k \times \omega_k(t)$$

where h , c and ω represent the PSF of the microscope, the constant molecular brightness and the time-dependent function of the fluctuating molecular brightness. As done originally by SOFI¹², which exploits the individual fluctuation character of each molecule, we can calculate the correlation cumulant for each pixel along the t axis to enhance the resolution. Taking the second-order temporal cumulant G_2 with zero time lag leads to

$$G_2(r) = \langle \delta F(r, t) \cdot \delta F(r, t) \rangle_t, \quad \delta F(r, t) = F(r, t) - \langle F(r, t) \rangle_t$$

where $\langle \cdot \rangle_t$ denotes time-averaging. Then, expanding the expression for G_2 , we can obtain

$$G_2(r) = \sum_{i,k} h(r - r_i) \times h(r - r_k) \times c_i \times c_k \times \langle \delta \omega_i(t) \cdot \delta \omega_k(t) \rangle_t$$

Assuming that each molecule is fluctuating individually (not correlating), the cross-correlation terms in this equation can be regarded as zero when $i \neq k$. Therefore, the second-order temporal cumulant can be written as the simple sum of the squared PSF weighted by the corresponding constant brightness γ :

$$G_2(r) = \sum_i h(r - r_i)^2 \times c_i^2 \times \langle \delta \omega_i(t)^2 \rangle_t = \sum_i h(r - r_i)^2 \times \gamma_i$$

The width of the resulting PSF in this equation is reduced by a factor of $\sqrt{2}$, so such a second-order temporal cumulant can improve the resolution by ~ 1.4 times at most.

Major steps of SACD reconstruction. In real experiments, the real fluorescent content we image includes not only in-focus fluorescent molecules, but also out-of-focus and cytosol fluorescence background and baseline signals (Extended Data Fig. 1a), which may not be in stationary equilibrium. The effective on/off contrast ratio is influenced by inessential fluorescence, and the effective SNR condition is dominated by the inevitable readout noise, dark current noise and shot noise of the sensor (Extended Data Fig. 1a), which will affect the accuracy of the cumulant calculation and the quality of the reconstructed image^{19,25,33}. The real fluorescent content is thus given by

$$F(r, t) = \sum_{k=1}^N h(r - r_k) \times [c_k \omega_k(t) + s_{r_k}(t)]$$

where s_{r_k} represents the mixed influence of the cytosol background and baseline noise signal at position r_k . Thus G_2 can be rewritten as

$$G_2(r) = \sum_{i,k} h(r - r_i) \times h(r - r_k) \times c_i \times c_k \times \langle [\delta \omega_i(t) + s_{r_i}(t)] \cdot [\delta \omega_k(t) + s_{r_k}(t)] \rangle_t$$

The cytosol background and baseline noise signal are usually assumed to be smoothly varying in the whole FOV, being spatiotemporally correlated. Thus, the cross-correlation terms in this equation cannot be treated as zero, which will compromise the resolution enhancement. Such spatiotemporally correlated cytosol and baseline

signals, as well as the noise components, can be reduced effectively by increasing the number of frames, as demonstrated in previous work¹².

Here we apply a pre-deconvolution step before the cumulant calculation as a tool to simultaneously reduce the noise, reject the out-of-focus and cytosol baseline signals, and improve the resolution. With this pre-deconvolution, the resolution enhancement ability of the temporal correlation cumulant can be applied more efficiently. Specifically, we use RL deconvolution^{27,28} without any regularization as the first step of our SACD reconstruction. Using a few iterations, the RL deconvolution can filter the noise and sharpen the image, without introducing noticeable artefacts. We used the following accelerated RL deconvolution here:

$$\begin{aligned} d^{j+1} &= d^j \cdot \left(h^T \cdot \frac{g}{h \cdot d^j} \right) \\ v^j &= d^{j+1} - y^j \\ \alpha^{j+1} &= \frac{\sum v^j \cdot v^{j-1}}{\sum v^{j-1} \cdot v^{j-1}} \\ y^{j+1} &= d^{j+1} + \alpha^{j+1} \times (d^{j+1} - d^j) \end{aligned}$$

where y^{n+1} is the image after $n+1$ iterations, g is the input image, and h is the PSF. The adaptive acceleration factor α was introduced by Andrews and colleagues³⁸; this represents the length of an iteration step, which can be estimated directly from the experimental results. We cautiously stop the iterations at a very early stage (usually ten iterations or a number automatically determined as in FRC-assisted SACD) to avoid destroying the independence of the fluorescence fluctuation.

To match the improved spatial resolution with the subsequent autocorrelation cumulant calculation, we apply Fourier interpolation²⁹ to the image stack after the pre-deconvolution step, which can lead to a smaller pixel size and larger pixel number, while keeping the real image content unchanged. We then calculate the autocorrelation cumulant of the RL-processed, Fourier-interpolated image sequence to improve the resolution. Considering that the resulting cumulant image is background-free, the deconvolution approach is a very suitable means with which to improve the image resolution further³⁰. In the final stage, a long RL deconvolution is used on the single image after autocorrelation reconstruction with the square of the PSF, which further extends the spatial resolution along the three dimensions. Overall, these major steps included in SACD reconstruction will achieve a more efficient twofold resolution enhancement, resulting in a massive reduction in the required sequence length.

Next, we follow the balanced SOFI (bSOFI)³⁰ configuration to linearize the intensity to rebuild the final SACD image. Because the intensity after the second-order cumulant will be scaled to its squared value, we took a square root of the image to linearize the brightness. To reduce the amplified noise induced by this gamma correction, small values (typically $\sim 2\%$ of the maximum value) are truncated, so the image is subsequently reconvolved using a small Gaussian kernel (with the FWHM as the corresponding doubled resolution).

FRC-assisted parameterless SACD

To make our SACD method easy to use, we use FRC to automatically determine the optimal iteration time (Extended Data Fig. 1b), as this has been used as an effective resolution criterion for SR and electron microscopy^{21,22,39}. The FRC calculation needs statistically independent image pairs that share identical details (correlated) but different noise realizations (uncorrelated). Because the FRC describes the highest acceptable frequency component of distance between two signals, we use it to automatically estimate a reliable cutoff frequency to determine the best iteration. We aim to obtain a higher reliable cutoff frequency, which indicates better image quality or resolution.

Specifically, we split the raw image sequence in half (odd and even frames) and subsequently average these subsets to generate the required two frames for the FRC calculation. We then estimate the FRC

value of each iteration result; in practice this is usually a quadratic curve along the iteration times. Although the number of iterations related to the maximum resolution (lowest FRC value) could be used (18 iterations in Extended Data Fig. 1c), we aim to be conservative in choosing the optimal iteration numbers for pre-deconvolution to avoid possible disturbance of the later autocorrelation. As shown in the representative example in Extended Data Fig. 1b, we generally stop the pre-RL deconvolution at half the iteration time for maximal resolution (nine iterations in Extended Data Fig. 1c). However, with no such concern in post-deconvolution, the iterative process could be stopped at the maximum FRC resolution reached.

In addition to determining the optimal iteration times, we can also utilize FRC to estimate the PSFs of the corresponding images (Extended Data Fig. 1b). In previous work, the global FRC value was used to estimate the PSF of the image, and this estimated PSF was applied in blind RL deconvolution²². However, it is notable that FRC estimates the effective PSF (the reliable frequency components), rather than the native resolution of the microscope system. The large noise amplitude will dominate the frequency components even inside the optical transfer function (OTF), which will influence the FRC to infer a larger resolution. On the other hand, the RL deconvolution requires the native resolution of the corresponding microscope system. Thus, this FRC-based PSF estimation introduces the assumption that the corresponding image is under sufficiently high SNR conditions, which is usually not met for raw images. Therefore, we calculate the PSF with the Born–Wolf model⁴⁰ in the pre-deconvolution step, and execute the PSF estimation using FRC on the noise-free autocorrelation reconstruction result in the post-deconvolution step.

Sparse-SACD reconstruction

Although the pre-RL deconvolution has been used in SACD to reduce the noise and background, it still has limits when dealing with images containing large noise and strong background signals. For sensitive live-cell SR imaging, the achievable SNR is severely limited by photobleaching and phototoxicity, and it will decrease continually during long-term image sequence acquisition. Predictably, the SACD-reconstructed result suffers from artefacts when the SNR is lowered to a certain level. This issue could be resolved by increasing the number of captured images. Furthermore, when the image contains a strong cytosol signal, sometimes even increasing the sequence length cannot fully remove this background. Remarkably, these two problems are common in fast and long-term live-cell imaging.

To maintain the temporal resolution of SACD, we applied the previously developed Sparse deconvolution²⁴ in our SACD reconstruction (Sparse-SACD) to address the above limitations. The Sparse deconvolution is a constrained deconvolution algorithm that can reduce artefacts and increase the spatial resolution. The framework contains two main steps, background subtraction and iterative deconvolution with sparsity and continuity constraints. Specifically, we use background subtraction on the raw images to remove the unnecessary low-frequency baseline signal and strong cytosol background (optional, before step 1 in SACD). The constrained iterative deconvolution is applied at the last step (replacing step 4 in SACD), which enables further resolution improvement while maintaining image quality (minimizing artefacts).

We use an iterative wavelet decomposition method to estimate and subtract the background. The background is iteratively estimated from the lowest-frequency wavelet bands of the images, and, in each iteration, all values of the image above the current estimated background level are clipped. The constrained iterative deconvolution is given by

$$\arg \min_y \left\{ \frac{\lambda}{2} \|G_2 - h^2 y\|_2^2 + \mathfrak{R}(y) \right\}$$

The first term on the left side of the equation is the fidelity term, representing the distance between the recovered image y and the result

obtained with the calculation of autocorrelation image G_2 . h is the PSF of the optical system. The second term represents the continuity and sparsity constraints, which is given by

$$\begin{aligned} \mathfrak{R}(y) = & \left\| \begin{matrix} y_{xx} & y_{xy} & \sqrt{\lambda_x} y_{xt} \\ y_{yx} & y_{yy} & \sqrt{\lambda_y} y_{yt} \\ \sqrt{\lambda_x} y_{tx} & \sqrt{\lambda_y} y_{ty} & \lambda_t y_{tt} \end{matrix} \right\|_1 + \lambda_{e1} \|y\|_1 \\ = & \|y_{xx}\|_1 + \|y_{yy}\|_1 + \lambda_t \|y_{tt}\|_1 + 2\|y_{xy}\|_1 + 2\sqrt{\lambda_x} \|y_{xt}\|_1 + 2\sqrt{\lambda_y} \|y_{yt}\|_1 + \lambda_{e1} \|y\|_1 \end{aligned}$$

The first term is the continuity prior regarding the structural continuity along the $xy-t(z)$ axes, and the second term is the sparsity constraint. λ_t denotes the regularization parameters that present the continuity along the t axis, which can be turned off upon imaging objects moving at high speed. The subscript xx , for example, is the two-order derivation operator in the x direction. λ and λ_{e1} denote weight factors to balance the image fidelity with sparsity.

The high-frequency artefacts caused by the readout noise, dark current and shot noise are mostly non-continuous along the $xy-t(z)$ axes. Moreover, to ensure sufficient Nyquist sampling criteria for maximal spatial resolution dictated by the optics, the PSF must occupy more than 3×3 pixels in space, which constitutes the basis for the continuity along the x and y axes. On the other hand, fluorescence microscopy achieves high resolution and contrast by labelling designated structures, which are always sparse in the spatial domain in terms of the existence of a fraction of nonzero value. Moreover, the fluorescence signal is blurred by the optical system, and it is accompanied by noise when captured by the photoelectric detector. Therefore, the real signal is always more sparse than the recorded signal²⁴. Accordingly, the continuity prior and the sparsity prior are involved in our Sparse-SACD to constrain the final step deconvolution to reduce the artefacts and improve the resolution. As described previously, to solve such a convex optimization problem, we split it into constraint reconstruction and deconvolution subproblems. We solve these two subproblems using a two-step operation, to gradually and effectively search for the final solution y . Specifically, the Split–Bregman framework⁴¹ is employed to execute the reconstruction of continuity and sparsity weighted constraints and we use RL deconvolution for the iterative deconvolution subproblem.

SD-confocal set-up

A commercial SD-confocal microscope system—a spinning-disk system (Dragonfly, Andor) based on an inverted fluorescence microscope (DMi8, Leica) with an oil immersion objective ($\times 100/1.3$ NA oil, Plan Apo, Leica)—was used in this work. Four laser beams (405 nm, 488 nm, 561 nm and 647 nm) were combined with the microscope. The images were captured either by an sCMOS camera (Zyla 4.2 Plus, Andor) or an electron-multiplying charge-coupled device camera (iXon Ultra 888, Andor).

SIM set-up

The SIM system was based on a commercial inverted fluorescence microscope (IX83, Olympus) equipped with an objective ($\times 100/1.7$ NA HI oil, APON, Olympus) and a multiband dichroic mirror (ZT405/488/561/640-phase R; Chroma) as described previously³⁴. In short, laser light with wavelengths of 488 nm (Sapphire 488LP-200) and 561 nm (Sapphire 561LP-200, Coherent) and acoustic optical tunable filters (AOTF, AA Opto-Electronic) were used to combine, switch and adjust the illumination power of the lasers. A collimating lens (focal length, 10 mm; Lightpath) was used to couple the lasers to a polarization-maintaining single-mode fibre (QPMJ-3AF3S, Oz Optics). The output lasers were then collimated by an objective lens (CFI Plan ApoChromat Lambda $\times 2/0.10$ NA, Nikon) and diffracted by a pure phase grating consisting of a polarizing beamsplitter, a half-wave plate and the

SLM (3DM-SXGA, ForthDD). The diffraction beams were then focused by another achromatic lens (AC508-250, Thorlabs) onto the intermediate pupil plane, where a carefully designed stop mask was placed to block the zero-order beam and other stray light and to permit passage of ± 1 ordered beam pairs only. To maximally modulate the illumination pattern while eliminating the switching time between different excitation polarizations, a home-made polarization rotator was placed after the stop mask³⁴. Next, the light passed through another lens (AC254-125, Thorlabs) and a tube lens (ITL200, Thorlabs) to focus on the back focal plane of the objective lens, which interfered with the image plane after passing through the objective lens. Emitted fluorescence collected by the same objective passed through a dichroic mirror, an emission filter and another tube lens. Finally, the emitted fluorescence was split by an image splitter (W-VIEW GEMINI, Hamamatsu) before being captured by an sCMOS camera (Flash 4.0 V3, Hamamatsu).

Wide-field and TIRF microscopy set-up

The three phases of structured illumination under the same orientation can be averaged to a uniform wide-field illumination. Taking advantage of this, we used the SIM set-up described in the previous section to generate wide-field images by integrating three frames (corresponding to three phases of structured illumination) on the camera plane, which enabled more flexible cross-validation of the SIM and SCD results. In other words, we used an identical commercial inverted fluorescence microscope (IX83, Olympus) equipped with an objective ($\times 100/1.7$ HI oil, APON, Olympus) and a sCMOS camera (Flash 4.0 V3, Hamamatsu) to capture the wide-field images for our SCD reconstruction.

High-throughput SR image acquisition and reconstruction

For high-throughput imaging of microtubules in COS-7 cells, we used a commercially available solution—the same type of SD-confocal microscope equipped with a montage stitching tool (Dragonfly, Andor)—to define the positions of 704 FOVs (each containing $1,024 \times 1,024$ pixels) of $2.0 \text{ mm} \times 1.4 \text{ mm}$ on a 32×22 grid, with 5% overlap. The stage was automatically shifted to each of the 704 positions. The exposure time at each FOV was ~ 30 ms (~ 40 ms including readout time), and the time series was 20 frames at each position. The total imaging time was ~ 9 min 23 s, and the stage moving time was ~ 10 min, so the total acquisition time was ~ 20 min.

The full stitched image stack ($30,193 \times 20,913 \times 20$ pixels) was cut to 759 (33×23) partly overlapping (12.5%) tiles, each $1,024 \times 1,024 \times 20$ pixels. We employed SCD reconstruction individually on these tiles, and the resulting SR images were subjected to histogram matching (with the previous tile as reference) to normalize the intensity for better subsequent fusion quality. Finally, the tiles were stitched back to a full SR image ($60,386 \times 41,826$), and this image was downsampled 12 times ($5,032 \times 3,486$ pixels) for the visualizations in Fig. 3a and Supplementary Fig. 3a.

Cell maintenance and preparation

COS-7 cells (ATCC, CRL-1651) were cultured in high-glucose Dulbecco's modified Eagle medium (DMEM; Gibco, 21063029) supplemented with 10% fetal bovine serum (FBS; Gibco) and 1% 100 mM sodium pyruvate solution (Sigma-Aldrich, S8636) in an incubator at 37°C with 5% CO_2 until $\sim 75\%$ confluency was reached. For the SD-confocal imaging experiments, 35-mm glass-bottomed dishes (Cellvis, D35-14-1-N) were used. For the wide-field and 2D-SIM imaging experiments, cells were seeded onto coverslips (H-LAF 10L glass; reflection index of 1.788; diameter of 26 mm; thickness of 0.15 mm, customized) coated with 0.01% poly-L-lysine solution (Sigma, P4707) for 10 min and washed twice with sterile water before seeding transfected cells.

Immunofluorescence

The COS-7 cells were grown in 35-mm glass-bottomed dishes overnight and rinsed with PBS, then immediately fixed with prewarmed

4% paraformaldehyde (PFA; Santa Cruz Biotechnology, sc-281692) for 10 min. After three washes with PBS, the cells were permeabilized with 0.1% Triton X-100 (Sigma-Aldrich, X-100) in PBS for 10 min. The cells were blocked in 5% BSA/PBS for 1 h at room temperature. Mouse anti-tubulin DM1a (Sigma, T6199) was diluted to 1:100 and stained cells in 2.5% BSA/PBS blocking solution for 2 h at room temperature. The cells were then washed with PBS five times for 10 min per wash and stained with biotin-XX goat anti-mouse immunoglobulin-G antibody (Invitrogen, B2763) at room temperature in PBS (1:200) for 1 h. The cells were then washed with PBS five times for 10 min per wash and stained with Qdot 525 streptavidin conjugate (Invitrogen, Q10143MP) or Qdot 605 streptavidin conjugate (Invitrogen, Q10103MP) at room temperature in PBS (1:50) for 1 h. Finally, the cells were washed five times with PBS and imaged. According to the Qdot properties, all types of Qdot were excited at 405 nm (numbers in Qdot 525 and Qdot 605 indicate emission wavelength).

Live-cell samples

To label the mitochondria/microtubules in live cells, COS-7 cells were transfected with Skylan-S-TOM20 and MAP4-Skylan-S¹⁹. To label the granular component (GC) and dense fibrillar component (DFC) in the nucleolus, we transfected the GC marker B23-rsFusionRed3⁴² (a red reversibly photoswitchable fluorescent protein) and the DFC marker Nop56-Skylan-S in COS-7 cells. The transfections were executed using Lipofectamine 2000 (Thermo Fisher Scientific, 11668019) according to the manufacturer's instructions. After transfection, cells were plated in glass-bottomed dishes. The Skylan-S was under sequential illumination with a 405-nm laser (low power) when imaging. In addition, live cells were imaged in a complete cell culture medium containing no phenol red in a 37°C live-cell imaging system.

Simulation protocol for the fluctuating fluorescence samples

Ground-truth structures were created on a fine grid (10 nm or 20 nm). Then, following simulation procedures in the SOFI simulation tool⁴³, the blinking behaviour of each emitter was modelled as a time-continuous Markovian process. Using that, we generated temporal blinking sequences of the two fluorescence emitters and then convolved these image sequences with a 220-nm PSF. The resulting images were then subsampled multiple times for a pixel size of 60 nm. For the 'Non-bias' condition in Extended Data Fig. 3a, only the Poisson noise was involved in the images. To synthesize more realistic experimental imaging conditions, cytosol background, out-of-focus light, Gaussian readout noise and baseline background under different levels ('low', 'medium' and 'high' in Extended Data Figs. 3b–d and 4 and Supplementary Fig. 6; 'ultralow' in Supplementary Fig. 13) were included sequentially in the images.

Performance metrics

FRC resolution. The calculation of FRC resolution requires two independent frames of identical content under the same imaging conditions²¹. For the case of SOFI and SCD, these two frames were generated by splitting the raw image sequence into two image subsets (the first 20 frames and the last 20 frames) and reconstructing them independently. For the wide-field and SIM cases (Fig. 1e), the raw image sequence was divided into two image subsets (the first 20 frames and the last 20 frames), then these subsets were averaged to create the required two frames.

Structural similarity. The structural similarity (SSIM)⁴⁴ is defined as

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + c_1)(2\sigma_{x,y} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}$$

where x represents the reference, which is ground-truth structures convolved with a 110-nm PSF, and y denotes the corresponding SR

reconstructions. μ_x and μ_y are the averages of x and y ; $\sigma_{x,y}$ is the covariance of x and y ; σ^2 are the variances; and c_1, c_2 are the variables used to stabilize the division with a small denominator.

Ring quality ratio. Because one two-peak analysis may not be sufficient to validate that the ring is fully resolved²⁴, to quantitatively evaluate the reconstruction quality we divided two circular/annular areas based on the corresponding ground-truth diameters. Following the modulation transfer function, we defined the ring quality ratio (RQR) as

$$\text{RQR} = 1 - \frac{\text{Avg}(\text{Region}_0)}{\text{Avg}(\text{Region}_1)}$$

where Avg calculates the mean intensity of pixels within the corresponding area. Region₀ and Region₁ represent the double annulus region (Extended Data Fig. 4h, in red) and the region in between (Extended Data Fig. 4h, in yellow). To avoid overconfident determination, we used a threshold of 0.4 to determine whether the ring structures are successfully resolved.

Standard deviation. To measure the reconstruction uncertainty (standard deviation (STD) of reconstruction) of fluctuation-based SR methods, following ref. 18, jackknife resampling was used to create a reconstruction stack. The resulting image stack then performed the STD calculation:

$$\text{STD} = \sum_{x,y} \sqrt{\frac{\sum_{i=1}^n (I_i(x,y) - \langle I(x,y) \rangle_{1 \sim n})^2}{n-1}}$$

where n is the sequence length; (x,y) denotes the 2D coordinate; $I_i(x,y)$ represents the intensity of the i th image in the sequence, and $\langle I(x,y) \rangle_{1 \sim n}$ is the averaged intensity of the sequence.

Reconstruction algorithms and their parameters for fluorescence fluctuation-based data

We compared performances against three other modified SOFI methods (SOFI-wavelet³², SOFI-RL^{30,31} and RD-Covar³³ (covariance calculation followed by RL deconvolution)) and five other freely available software packages (SRRF¹³, ESI¹⁵, HAWK⁴⁵, ThunderSTORM⁴⁶ and bSOFI³⁰). Notably, because the three modified SOFI methods are not open-sourced, we implemented them with adherence to the given published works. We will describe how we configured each of these methods in turn.

SOFI. A 2D Fourier upsampling operation was performed on the raw stack, followed by a second-order cumulant calculation.

SOFI-RL. A 2D Fourier upsampling operation was performed on the raw stack, followed by a second-order cumulant calculation. We then applied RL deconvolution to the resulting image.

SOFI-wavelet. A 2D Fourier upsampling operation was performed on the raw stack, followed by a two-step wavelet-based filter. The first step is equal to a high-pass filter along the t axis for each pixel to remove background fluctuations, and we used 2D Daubechies-4 wavelet filters to decompose the signal up to the seventh level and remove the lowest component. The second step is similar to a low-pass filter on each image to moderate readout noise, and we used 2D Daubechies-4 wavelet filters to decompose the signal up to the fourth level and cut off the highest component. After that, the second-order cumulant was employed on the image stack.

RD-Covar. A 2D Fourier upsampling operation was performed on the raw stack, followed by a second-order covariance calculation, then

we applied RL deconvolution to the resulting image. The covariance calculation is defined as

$$\text{Cov}_2 = -\frac{(N-2) \cdot M_{1,1}^2}{N \cdot (N-1)} + \frac{M_{0,2} \cdot M_{2,0}}{N \cdot (N-1)} + \frac{M_{2,2}}{N}$$

where $M_{m,n} = \langle \delta F^m(\mathbf{r}, t) \cdot \delta F^n(\mathbf{r}, t + \tau) \rangle$, $\delta F(\mathbf{r}, t) = F(\mathbf{r}, t) - \langle F(\mathbf{r}, t) \rangle$. In specific, τ is a time lag and set to be 0 in this work, $F(\mathbf{r}, t)$ represents the recorded signal intensity at position $\mathbf{r}$ and time t , and N is the sequence length.

SRRF ImageJ plugin. Images labelled with ‘SRRF’ were reconstructed by the temporal radially average method, with default parameters, except with the ‘Radiality magnification’ as ‘2’ for Fig. 1c and ‘4’ for Supplementary Fig. 12. Images labelled with ‘SRRF-TRC’ and ‘SRRF-TRAC’ in Extended Data Fig. 8 were reconstructed by the temporal radially average and the temporal radially autocorrelation methods, respectively, with default parameters, except with the ‘Radiality magnification’ set as ‘2’.

ESI ImageJ plugin. Images labelled with ‘ESI-2nd’ in Fig. 1c and Extended Data Fig. 8 and ‘ESI’ in Supplementary Fig. 5 were reconstructed by second-order and fourth-order ESI respectively, with default parameters.

HAWK ImageJ plugin. The image labelled with ‘HAWK-SOFI’ in Extended Data Fig. 8 was reconstructed by HAWK preprocessing with default parameters followed by SOFI.

ThunderSTORM ImageJ plugin. The image labelled with ‘ThunderSTORM’ in Supplementary Fig. 5 was reconstructed by maximum likelihood fitting with the integrated PSF method and multi-emitter fitting enabled. Drift correction was performed post-localization and images were rendered using an average shifted histogram approach.

bSOFI MATLAB library. The image labelled with ‘bSOFI’ in Supplementary Fig. 5 was reconstructed by the fourth-order bSOFI with default parameters.

Image rendering and processing

Diffraction-limit image rendering. We applied a rolling ball background subtraction on the diffraction-limited image and filtered the resulting image with a small Gaussian kernel (1 pixel in radius) to remove readout noise.

SOFI image rendering. We followed the bSOFI³⁰ configuration to visualize the SOFI images. Notably, to avoid cancelling the resolution improvement of the cumulant, the SOFI images without post-deconvolution were directly visualized by the raw version with brightness adjustment.

Colour map and software. The colour map ‘biop-12colors’ was used to colour-code the 3D volumes in Figs. 2e and 4f. The colour map ‘Isolum’⁴⁷ was used to colour-code the 3D volume in Extended Data Fig. 7a. The colour map ‘SQUIRREL-FRC’⁴⁸ was used to colour-code the FRC maps in Fig. 3a and Supplementary Fig. 3e. The colour map ‘Morgenstemning’⁴⁷ was applied to show the Fourier-transform results in Extended Data Fig. 5c. The colour map ‘SQUIRREL-errors’⁴⁸ was used to colour-code the resolution-enhancing map in Supplementary Fig. 3e. The colour map ‘Jet’ was used to show the time series in Supplementary Fig. 12h, and the intensity amplitudes in Extended Data Fig. 10b. All data processing was achieved using MATLAB and ImageJ. All figures were prepared with MATLAB, ImageJ, Microsoft Visio and OriginPro, and all videos were produced with our lightweight MATLAB framework, which is available at <https://github.com/WeisongZhao/img2vid>.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this Article.

Data availability

Several representative datasets are available at <https://github.com/WeisongZhao/SACDj>. All the data that support the findings of this study are available from the corresponding author on request.

Code availability

The SADC in the MATLAB library can be found at <https://github.com/WeisongZhao/SACDm>, and its corresponding ImageJ plugin is available at <https://github.com/WeisongZhao/SACDj>.

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Author contributions

H.L., C.G., L.C. and W.Z. supervised the project. W.Z. and H.L. initiated and conceived the research. W.Z. developed the algorithm and implemented the corresponding software. W.Z. and S.Z. designed the experiments. S.Z. performed the experiments and collected the data. W.Z. analysed the data and prepared the figures with contributions from Xiangyan Ding. W.Z. performed the simulations with contributions from Z.H., Xiangyan Ding, L.Q., Y.H. and X.W. Z.H. and W.Z. prepared the videos. H.M., Y.J., Y.Hu, Xumin Ding, G.H., C.G. and J.T. participated in discussions during the development of the manuscript. W.Z., H.L. and L.C. wrote the manuscript with input from all authors. All authors participated in the discussions and data interpretation.

Competing interests

L.C., H.L., W.Z. and S.Z. have a pending patent application on the presented framework.

Additional information

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Correspondence and requests for materials should be addressed to Xumin Ding, Liangyi Chen, Changliang Guo or Haoyu Li.

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